Linear Algebra - Section 1.4
The Matrix Equation

→ We can view linear combinations as the product of a matrix and a vector.

- Def: Let A be an $n \times n$ matrix w/ columns $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$. Let $\vec{x}$ be a vector in $\mathbb{R}^n$, then $A\vec{x}$ is the linear combination of the columns of A w/ $\vec{x}$. That is,

$$A\vec{x} = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

Note that the # of columns = # of entries in $\vec{x}$.

- Ex: $\begin{bmatrix} 2 & -3 & -5 \\ 0 & -5 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \\ 3 \end{bmatrix}$

$$= 4\begin{bmatrix} 2 \\ 0 \end{bmatrix} + 7\begin{bmatrix} -3 \\ -5 \end{bmatrix} + 3\begin{bmatrix} -5 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 \\ 0 \end{bmatrix} + \begin{bmatrix} -21 \\ -35 \end{bmatrix} + \begin{bmatrix} -15 \\ -3 \end{bmatrix} = \begin{bmatrix} -28 \\ -38 \end{bmatrix}.$$

- If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 10 \\ 11 \\ 12 \end{bmatrix}$

then we can write this as a matrix equation OR a vector equation:

same but different

matrix equation: $A\vec{x} = \vec{b}$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 11 \\ 12 \end{bmatrix}$$

vector equation: $x_1\vec{a_1} + x_2\vec{a_2} + x_3\vec{a_3} = \vec{b}$

$$x_1 \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 10 \\ 11 \\ 12 \end{bmatrix}$$

- Theorem: (Existence of Solutions)
  The matrix equation $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is a linear combination of the columns of A.